

ESQUEL, Argentina

WMO: 878030

Lat: 42.9058S

Lon: 71.1450W

Elev: 797

StdP: 92.11

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-7.5	-5.4	-10.9	1.6	0.4	-9.0	1.9	2.1	19.5	8.1	15.7	7.4	1.4	20	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	27.8	14.7	25.6	13.7	23.6	12.8	15.4	25.7	14.4	24.0	13.4	22.3	5.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
10.7	8.8	18.4	9.7	8.2	17.9	8.7	7.7	17.1	45.6	26.1	42.7	24.1	40.0	22.3	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	14.3	12.5	-11.3	31.2	3.1	1.6	-13.5	32.3	-15.3	33.3	-17.0	34.2	-19.3	35.4		
(5)			-11.7	17.3	3.0	1.3	-13.9	18.2	-15.6	19.0	-17.3	19.7	-19.5	20.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	15.2	15.1	12.6	8.9	5.8	3.1	2.4	3.9	6.0	8.6	11.1	13.7		
	DBStd	5.54	3.53	3.65	3.20	3.15	3.37	3.57	3.78	2.95	2.86	2.85	3.07	3.36		
	HDD10.0	1059	3	3	11	57	136	209	236	191	123	62	22	7		
	HDD18.3	3502	110	103	178	284	389	458	494	448	371	301	217	148		
	CDD10.0	629	163	145	92	23	5	1	1	1	3	19	56	121		
	CDD18.3	29	12	12	1	0	0	0	0	0	0	0	1	4		
	CDH23.3	482	166	179	37	0	0	0	0	0	0	2	19	79		
	CDH26.7	106	37	47	5	0	0	0	0	0	0	0	2	14		
(14) Wind	WSAvg	5.2	6.7	5.4	5.0	4.6	4.1	4.2	4.1	4.4	4.8	5.6	6.5	6.7		
(15) Precipitation	PrecAvg	473	22	18	21	41	70	81	68	55	32	29	21	23		
	PrecMax	707	123	60	72	164	176	214	187	159	116	105	59	101		
	PrecMin	262	0	0	3	2	2	10	3	13	1	0	0	0		
	PrecStd	121	24	18	15	34	47	49	43	32	26	24	15	24		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	30.8	27.1	21.8	16.8	13.5	13.4	15.1	18.3	22.2	26.1	28.6		
		MCWB	15.9	16.1	14.5	12.3	10.1	7.9	7.2	8.4	8.7	10.8	13.0	15.4		
	2%	DB	27.4	28.0	23.9	18.7	14.8	11.3	10.8	12.3	15.4	19.2	22.5	25.4		
		MCWB	14.5	14.7	13.4	10.9	9.2	6.7	6.0	6.9	7.6	9.6	11.5	13.5		
	5%	DB	25.0	25.4	21.8	16.8	13.0	9.9	9.2	10.6	13.4	16.8	20.0	23.2		
		MCWB	13.6	13.7	12.3	9.7	8.2	5.7	5.2	5.8	6.6	8.4	10.4	12.2		
	10%	DB	22.8	23.0	19.4	15.0	11.3	8.4	7.7	9.0	11.6	14.8	17.7	20.9		
		MCWB	12.5	12.8	11.5	9.0	7.0	4.7	4.1	4.8	5.7	7.6	9.4	11.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	17.1	15.6	13.1	11.1	8.8	8.0	9.3	9.6	12.0	13.7	16.4		
		MCDB	28.5	29.0	25.2	19.9	15.3	12.6	11.4	14.0	16.6	20.2	24.3	27.2		
	2%	WB	15.2	15.3	14.1	11.5	9.7	7.2	6.5	7.5	8.2	10.2	12.1	14.2		
		MCDB	25.7	25.5	22.4	16.9	13.9	10.6	10.0	11.2	14.2	17.7	21.0	24.1		
	5%	WB	14.0	14.3	13.0	10.4	8.5	6.1	5.5	6.3	7.1	9.1	10.9	12.9		
		MCDB	23.7	24.0	20.6	15.4	12.3	9.3	8.6	9.8	12.3	15.9	18.9	21.8		
	10%	WB	13.0	13.2	11.9	9.4	7.5	5.0	4.4	5.1	6.1	8.0	9.7	11.8		
		MCDB	21.5	21.9	18.6	14.3	10.8	7.7	7.2	8.4	10.8	13.8	16.9	20.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	11.9	11.2	10.1	8.3	7.0	7.3	8.1	9.7	10.3	10.5	10.7		
		MCDBR	16.1	16.6	15.9	13.6	10.4	8.3	8.8	10.5	13.2	15.1	15.2	15.8		
		MCWBR	6.8	7.0	7.7	7.4	6.4	5.5	5.4	6.2	7.1	7.6	7.3	6.9		
		MCDBR	14.3	14.9	14.0	11.6	9.2	8.0	8.1	9.2	11.4	13.5	13.4	14.2		
	5% WB	MCWBR	6.6	6.8	7.2	6.8	6.0	5.6	5.3	5.9	6.5	7.3	6.8	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.298	0.302	0.291	0.282	0.275	0.264	0.263	0.277	0.295	0.297	0.296	0.302		
	taud		2.587	2.568	2.601	2.616	2.593	2.576	2.562	2.566	2.522	2.537	2.561	2.552		
	Ebn at Noon		1030	998	966	906	839	815	843	897	945	992	1025	1031		
	Edh at Noon		101	98	86	72	61	56	61	74	91	100	103	106		
(44) All-Sky Solar Radiation	RadAvg		7.64	6.62	4.82	3.13	1.84	1.35	1.59	2.36	3.74	5.24	6.79	7.60		
	RadStd		0.39	0.39	0.32	0.26	0.21	0.16	0.15	0.23	0.33	0.39	0.40	0.53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+0.85	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	+0.86	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page